SUB-SAHARAN AFRICA

Health Indicators Business Analysis Report

Excel PivotTables • Indicator comparisons • 2000–2024 dataset period

This report summarizes the health, demographic, and socioeconomic patterns visible in the supplied Excel PivotTables. It focuses on comparisons across countries and on changes reported for selected indicators over the 2000–2024 period.

Main message: the PivotTables show substantial differences between countries in life expectancy, health expenditure, poverty, immunization, nutrition, and the balance of communicable and non-communicable deaths. A single regional average would hide important differences.

Executive summary

The supplied summaries point to several priorities for further investigation. Average life expectancy is highest among the countries listed for Seychelles (76.30 years), Cabo Verde (76.22), and Mauritius (73.79). The table also reports increases in life expectancy over the period for multiple countries, including Uganda (+18.85%), Botswana (+17.03%), and Eritrea (+12.60%). These are changes shown in the PivotTables, not independently recalculated estimates.

The poverty summary shows high average poverty rates for the Democratic Republic of the Congo (86.37), Mozambique (81.22), and Burundi (77.17). The undernourishment summary reports 63.76 for Somalia, 41.13 for Zambia, and 35.85 for Liberia. These figures identify areas for closer attention, but the report does not assume that they share the same measurement scale or reference period beyond the labels shown in the workbook.

The immunization summary shows comparatively high averages for the three measures listed for Seychelles, Mauritius, Botswana, and Cabo Verde. Meanwhile, the communicable/non-communicable death summaries suggest that countries differ in the relative contribution of these causes. These patterns can guide questions for further analysis, but they do not by themselves explain why the differences exist.

Purpose and intended audience

The report is written as a portfolio-style analytical brief for readers interested in data analysis, public health planning, and evidence-based decision-making. It demonstrates how Excel PivotTables can turn a large dataset into comparisons that are easier to interpret.

1. Key findings

Life expectancy

Country	Average life expectancy	Reported change
Seychelles	76.30 years	+3.52%
Cabo Verde	76.22 years	+8.10%
Mauritius	73.79 years	+2.13%
Uganda	68.49 years	+18.85%
Botswana	69.29 years	+17.03%
Eritrea	68.89 years	+12.60%

Interpretation: the displayed averages are highest for Seychelles and Cabo Verde among the countries visible in the summary. The percentage-change column also reports increases for Uganda, Botswana, and Eritrea. A useful next step would be to compare the starting and ending values directly and check whether each country's data coverage is complete for all years.

Poverty and undernourishment

Indicator	Country	Value shown in PivotTable
Average poverty rate	Congo, Dem. Rep.	86.37
Average poverty rate	Mozambique	81.22
Average poverty rate	Burundi	77.17
Average undernourishment	Somalia, Fed. Rep.	63.76
Average undernourishment	Zambia	41.13
Average undernourishment	Liberia	35.85

Interpretation: these values indicate substantial socioeconomic challenges in the listed countries. They support further investigation into how poverty and undernourishment vary over time and how those patterns relate to other indicators. The report does not infer a causal relationship from the summary tables alone.

Immunization

Country	Measles average	Polio (POL3) average	DPT average
Seychelles	97.64	97.84	97.84
Mauritius	95.92	96.32	95.92
Botswana	94.76	95.72	95.16
Cabo Verde	93.68	94.72	94.32
Eritrea	92.20	94.24	93.76

Interpretation: the selected country averages are high across the three immunization measures shown. However, averages can hide changes between years and differences within countries. Year-by-year trends and coverage gaps should be checked before drawing operational conclusions.

2. Health-system and disease indicators

Health expenditure

Country	Average health expenditure (% of GDP)
Namibia	9.65
Lesotho	8.72
Burundi	7.60
Liberia	7.56
South Africa	7.44

The PivotTable lists Namibia, Lesotho, Burundi, Liberia, and South Africa among the higher values visible for average health expenditure as a share of GDP. This percentage is not the same as spending per person or the absolute size of a health budget. To compare the resources available to health systems fairly, these measures should be considered separately.

Communicable and non-communicable deaths

Measure	Countries with higher values in the visible summary
Communicable-death share (average)	Niger 70.33; Chad 70.08; Nigeria 69.49
Non-communicable-death share (average)	Mauritius 84.48; Seychelles 77.54; Cabo Verde 63.27

Interpretation: the visible summaries suggest different cause-of-death profiles across countries. Countries with a higher communicable-death share may face a different mix of public-health needs from countries where non-communicable deaths account for a larger share. The figures are averages from the workbook; they should not be interpreted as counts of deaths, and the exact indicator definitions should be confirmed against the source dataset.

Tuberculosis and HIV incidence

The PivotTables list South Africa (921.28), Eswatini (920.28), and Lesotho (908.16) among the higher average TB-incidence values shown. For HIV incidence, the visible summary lists Eswatini (13.824), Lesotho (10.56), and Botswana (7.648). The units and denominators are not visible in the supplied PDF extract, so these figures are reproduced as displayed and should not be compared with external rates until their definitions are verified.

Safe water and smoking

The safe-water summary shows South Africa at 67.91, Botswana at 65.05, and Seychelles at 63.30. The smoking summaries list female averages of 23.55 for Madagascar and 22.10 for Comoros, while the male summary lists 48.85 for Madagascar and 45.05 for Lesotho. These results may help identify indicators for deeper analysis, but their units, age ranges, and measurement definitions should be checked before making direct comparisons.

3. Demographic and socioeconomic context

Population and urbanization

The population PivotTable shows the largest displayed multi-year sums for Nigeria, Ethiopia, the Democratic Republic of the Congo, and South Africa. Because the column is labelled 'Sum of population', these figures appear to add population values across multiple years; they should not be presented as current population estimates. For a meaningful country ranking, use the population value for a single selected year or calculate a clearly defined annual average.

The urbanization table reports average urban percentages of 86.58 for Gabon, 64.56 for Cabo Verde, 64.11 for Sao Tome and Principe, 62.58 for Botswana, and 61.71 for South Africa. These figures indicate variation in the urban share of the population among the listed countries. Urbanization can affect how health services are accessed, but this table alone does not show whether urbanization improves health outcomes.

Birth and death indicators

The workbook includes birth-rate and death-rate summaries as well as total births and total deaths. The visible percentage-change notes report decreases in birth rate for several countries, including Nigeria (25.97%), Ethiopia (30.97%), the Democratic Republic of the Congo (9.74%), and South Africa (14.31%). Death-rate changes are also shown for several countries, including Nigeria (-33.25% as labelled a decrease), Ethiopia (-59.96%), and South Africa (-11.12%). These changes should be checked against the underlying first- and last-year values before being used in a formal policy conclusion.

GDP indicators

The GDP per capita and GDP per capita PPP tables show high multi-year sums for some countries, including Seychelles and Equatorial Guinea. Because the tables use sums, these values are not equivalent to a single-year GDP per capita ranking. For cross-country comparison, select a specific year or use an appropriate average over the period, and clearly label the currency and price basis.

> **Data interpretation note: a PivotTable aggregation is part of the finding. A sum across years, an average across years, a percentage change, and a single-year value answer different questions and should not be treated as interchangeable.**

4. Recommendations

Based on the patterns visible in the supplied PivotTables, the following are practical next steps for analysis:

1. Prioritize year-by-year trend charts. Show the first and last year, the direction of change, and any years with missing data for each selected indicator.

2. Investigate high poverty and undernourishment values. Compare these indicators with life expectancy, safe-water access, and health expenditure, without assuming one causes another.

3. Review immunization coverage over time. Country averages can look strong even when there are gaps in individual years or populations.

4. Separate health spending measures. Report expenditure as a share of GDP separately from per-capita spending and total expenditure.

5. Use consistent units and definitions. Confirm whether disease-incidence figures are rates per population, percentages, or another measure before comparing countries.

6. Improve dashboard readability. Use one chart per question, clear titles and units, sorted categories, readable labels, and avoid placing many unrelated indicators on one chart.

5. Limitations and data-quality checks

This report is based on the supplied one-page PDF export of the Excel PivotTables. Some charts and labels are difficult to read because the columns and chart areas were expanded, as noted by the analyst. The report therefore uses values that are legible in the PivotTable text and does not estimate values from unclear chart shapes.

• **Coverage:** The report cannot confirm that every country has complete data for every year from 2000 through 2024.

• **Aggregation:** Several tables show sums across years. Multi-year sums of population, GDP per capita, or per-capita expenditure should not be described as current-year values.

• **Definitions and units:** The source definitions and denominators for some indicators are not visible in the PDF. In particular, disease-incidence and smoking values should be verified before comparison.

• **Change calculations:** Percentage changes are reproduced as labelled in the PivotTables. The original values and formulas were not available for independent recalculation.

• **Association versus causation:** Country-level comparisons can reveal patterns but cannot establish that one factor causes another.

• **Comparability:** Country averages can mask differences between regions, population groups, and years.

Conclusion

The Excel PivotTables provide a useful starting point for comparing health and socioeconomic indicators across Sub-Saharan African countries over the 2000–2024 period. The summaries highlight variation in life expectancy, poverty, undernourishment, immunization, health expenditure, and causes of death. They also show why data preparation and careful interpretation matter: averages, sums, and percentage changes must be clearly distinguished before insights are communicated.

The strongest next step is to refine the Excel dashboard around a small set of clearly defined questions, validate the units and aggregations, and use the findings to support focused recommendations. This would make the project more useful to decision-makers and demonstrate the full analytical process—from data summarization to interpretation and communication.

Source and method

Source: user-provided *health_indicators.pdf*, containing Excel PivotTables and charts for health and socioeconomic indicators. The report summarizes only information legible in that supplied file. No external sources were used, and figures have not been independently recalculated from the underlying dataset.